

Feature	LLM (Large Language Model)	Traditional ML Model
Definition	A deep learning model trained on massive text datasets to understand and generate human language.	A model trained on structured or labeled data to perform specific prediction or classification tasks.
Data Type	Primarily unstructured text (can also handle images, audio, code in multimodal models).	Usually structured/tabular, numerical, categorical, or domain-specific data.
Training Data Size	Billions to trillions of tokens.	Thousands to millions of records.
Training Cost	Very high (requires large GPU/TPU clusters).	Relatively lower.
Generalization	Can perform many tasks without retraining (translation, summarization, coding, Q&A, etc.).	Usually designed for a single task.
Examples	GPT-5, Llama, Gemini	Linear Regression, Decision Trees, Random Forest, XGBoost, SVM
Input	Natural language prompts.	Feature vectors or engineered inputs.
Output	Text, code, summaries, answers, reasoning, content generation.	Predictions, classifications, probabilities, recommendations.
Feature Engineering	Minimal; learns representations automatically.	Often requires manual feature engineering.
Transfer Learning	Strong; can adapt to many tasks via prompting or fine-tuning.	Limited; retraining is usually needed for new tasks.
Interpretability	Lower; difficult to explain internal reasoning.	Often higher, especially for simpler models.
Inference Cost	Higher computational requirements.	Usually faster and cheaper.
Typical Use Cases	Chatbots, virtual assistants, content generation, code generation, document analysis.	Fraud detection, sales forecasting, churn prediction, recommendation systems, demand forecasting.
Accuracy on Structured Data	Good, but not always optimal.	Often better for tabular business data.
Deployment Size	Usually GBs to TBs.	Usually MBs to a few GBs.